

Summary Pages for Peskin's Standard Model

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1 Chapter 3: Relativistic Wave Equations

1.1 Definitions

1.2 Summary

Introduces Lorentz invariance and relevant QFT aspects. Theories: KG, Dirac, QED, MW. Discusses the creating and annihilation of states, the relativistic normalisation of states and spin statistics.

1.3 Key Formulae

Creation of a state and its conjugate (Annihilation)

$$\left\langle 0 \left| \phi(x^\mu) \right| \varphi(p_\mu) \right\rangle = e^{-ip_\mu x^\mu} \quad \left\langle \varphi(p_\mu) \left| \psi^\dagger(x^\mu) \right| 0 \right\rangle = e^{+ip_\mu x^\mu} \quad (1.1)$$

We have the various theories

$$\begin{aligned} \mathcal{L}_{KG} &= \frac{1}{2} [\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2] & \implies & [\partial_\mu \partial^\mu + m^2] \phi = 0 \\ \mathcal{L}_{MW} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu & \implies & \partial_\mu F^{\mu\nu} = -J^\nu \\ \mathcal{L}_D &= \bar{\psi} [i\gamma^\mu \partial_\mu - m] \psi & \implies & [i\gamma^\mu \partial_\mu - m] \psi = 0 \\ \mathcal{L}_{QED} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} [i\gamma^\mu \partial_\mu - m] \psi. \end{aligned}$$

For relativistic normalisation we have

$$\left\langle p_1 \left| p_2 \right\rangle = 2E_{p_1} (2\pi)^3 \delta^{(3)}(\vec{p}_1 - \vec{p}_2) \quad (1.2)$$

1.4 Key Plot and Experimental Signature

1.5 Outline of a Paramount Derivation

Use normalisation of states to show

$$1 = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} |p\rangle \langle p| \quad (1.3)$$

1.6 Assumptions and Limiting Cases

1.7 Miscellaneous

2 Chapter 4: The Hydrogen Atom and Positronium

2.1 Definitions

Definition 2.1.1 (Positronium). The bound state of an electron–positron pair.

Definition 2.1.2 (Spin–Orbit Interaction). The splitting of the degeneracy of an electron’s states which have $\ell \neq 0$. e^- has spin $\frac{1}{2}$, $\implies \exists 2$ spin states. At a fixed (n, ℓ) , $\exists 2(2\ell + 1)$ states available for splitting.

Definition 2.1.3 (Fine Structure of the Hydrogen Atom). Splitting of the energy levels of hydrogen due to the spin–orbit interaction.

It takes order α^2 relative to the Rydberg energy and lowers the energy of states with smaller J and raises those with larger J .

Definition 2.1.4 (Hyper–Fine structure). The correction to the Fine Structure when we consider the spin of the proton inside the hydrogen atom.

It is smaller than the fine structure by 10^{-3} due to the large mass of the proton.

2.2 Summary

2.3 Key Formulae

Magnetic moment of an electron

$$\vec{\mu}_e = g \frac{Q}{2M} \vec{L} \quad (2.1)$$

Thomass Precession

$$\vec{\omega} = \frac{\gamma^2}{\gamma + 1} \vec{\alpha} \times \vec{v} \quad (2.2)$$

Spin $\times$ x

$$\begin{aligned} \vec{J} &= \vec{L} + \vec{S} \\ \vec{L} \cdot \vec{S} &= \frac{1}{2} [J^2 - L^2 - S^2]. \end{aligned}$$

2.4 Key Plot and Experimental Signature

2.5 Outline of a Paramount Derivation

Thomass Precession.

2.6 Assumptions and Limiting Cases

2.7 Miscellaneous

3 Chapter 5: The Quark Model

3.1 Definitions

Definition 3.1.1 (Hardons). Particles which are made of quarks.

Quarks never isolated, only constituents of Hadrons; thus strong interactions are done at the level of hadrons.

Definition 3.1.2 (Baryons). A subset of Hadrons which contain three quarks (qqq).

Examples include protons and neutrons. Each baryon has a baryon number of 1 while, non-baryons have baryon number equal to 0, baryon number is conserved. Proton is lightest baryon and thus by baryon number conservation they are stable particles.

Definition 3.1.3 (Mesons). A subset of Hardons which contain two quarks ($q\bar{q}$)

Examples include pions, kaons and charmonium $\frac{J}{\psi}$.

The lightest meson is the $\pi^0 \sim 135 MeV$

Definition 3.1.4 (Gluon). The spin 1 (boson) which mediates the strong force.

Definition 3.1.5 (Strangeness). K and K^* are only ever produced together, not in isolation. This, K mesons and a heavy excited state of a proton (Λ^0 baryons) are called the *strange particles*. The quantum number *strangeness* is preserved by the strong interaction. K^0, K^+, K^{*0} and K^{*+} have strangeness $S = -1$, their anti-particles and Λ^0 have $S = +1$.

Definition 3.1.6 (Isospin). A continuous symmetry in which the nuclear forces are invariant under.

Proton and neutron 2-component system, with transformations as $SU(2)$.

$$\begin{bmatrix} u \\ d \end{bmatrix} = e^{-i\vec{\alpha} \cdot \vec{\sigma} \cdot \frac{1}{2}} \quad (3.1)$$

The combinations $\bar{u}u, \bar{u}d, \bar{d}u, \bar{d}d$ form the $\frac{1}{2} \otimes \frac{1}{2}$ representation, (a sum of spin 0 and spin 1). The isospin 1 representation has three degenerate states, identifiable with three π and three ρ mesons. The isospin 0 states are identifiable with the η and ω mesons.

Definition 3.1.7 (G-Parity).

$$G := C e^{i\pi I_2} \quad (3.2)$$

A rotation of 180° around the $\hat{2}$ direction followed by charge conjugation.

Definition 3.1.8 (Charmonium). The meson which contains $c\bar{c}$ quarks.

Narrow resonance indicates long lifetime

Definition 3.1.9 (Bottomonium). The meson Υ comprised of $b\bar{b}$ quarks.

Definition 3.1.10 (Flavour). The flavour of quark is the type of quark, (a free index running over u,d,c,s,t,b)

3.2 Summary

Quark confinement to their bound states stem from potentials of the form $V(r) = A \log r$ or $V(r) = -\frac{A}{r} + Br$ agree with energies of states of the ψ and Υ .

3.3 Key Formulae

Lifetime of particle

$$\tau = \frac{\hbar}{\Gamma} \quad (3.3)$$

Isospin

$$[I_i, I_j] = i\varepsilon_{ijk}I_k \quad I_i = \frac{\sigma_i}{2} \quad (3.4)$$

Total width of a resonance Γ_A

$$\Gamma_A = \sum_f \Gamma(A \rightarrow f) \quad (3.5)$$

Branching ration (probability that a particular decay of A gives final state f)

$$BR(A \rightarrow f) = \frac{\Gamma(A \rightarrow f)}{\Gamma_A} \quad (3.6)$$

3.4 Key Plot and Experimental Signature

A *bubble chamber* is a volume of liquid in which moving charged particles create bubbles.

3.5 Outline of a Paramount Derivation

3.6 Assumptions and Limiting Cases

3.7 Miscellaneous

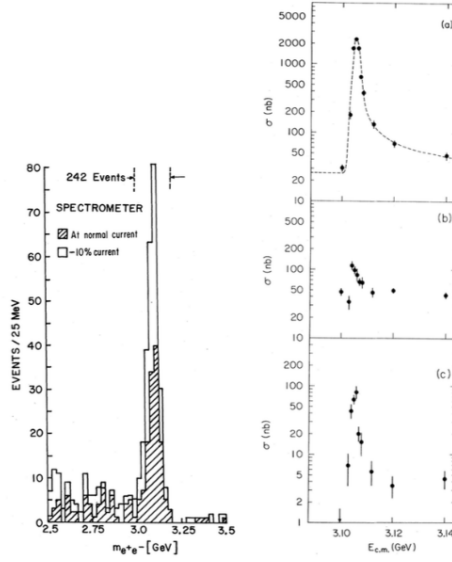


Figure 3.1: Narrow resonance at 3.1 GeV $\Rightarrow$ long lifetime, called $\frac{J}{\psi}$ - Charmonium. It is the bound state of $c\bar{c}$ which is found in $pp \rightarrow e^+e^- + X$. a- multiple hadrons, b- e^+e^- , c- pairs of μ , π , K

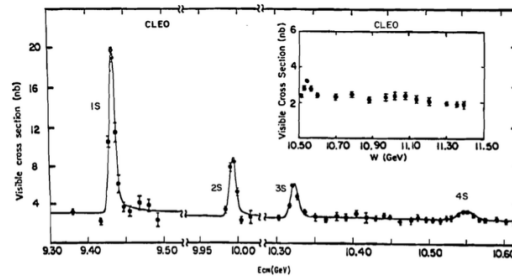


Figure 3.2: Resonances at 9460 MeV, Υ Bottomonium. Bound state of $b\bar{b}$ which is found in $pp \rightarrow \mu^+\mu^- + X$.

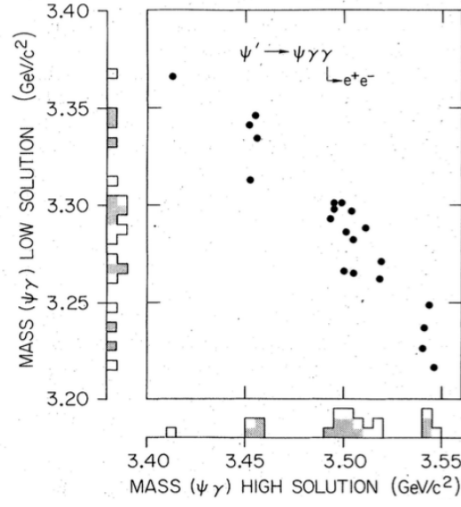


Figure 3.3: Each event contains $J/\psi + \gamma_1 + \gamma_2$. Since there exist 2 photons two invariant masses can form, $m\left(\frac{J}{\psi} + \gamma_1\right)$ or $m\left(\frac{J}{\psi} + \gamma_2\right)$. The photons come from excited states moving to lower energies $\psi' \rightarrow \gamma_1 \chi$ and $\chi \rightarrow \gamma_2 \frac{J}{\psi}$. Photons identical $\implies \exists$ ambiguity. X axis is "high solution" largest of $m\left(\frac{J}{\psi} \gamma\right)$ value. the three concentrations χ_{cJ} correspond to different values of angular momentum ($\chi_{c0} \rightarrow 3.41$), ($\chi_{c1} \rightarrow 3.51$) and ($\chi_{c2} \rightarrow 3.56$) all GeV.

Low energy photons (γ_1) have a narrower energy spread since ψ' is essentially at rest so all γ_1 's will give $E_{\gamma_1} = \frac{m_{\psi'}^2 - m_{\chi}^2}{m_{\psi'}}$, the value of γ_2 depends which direction it was emitted in $E_{\gamma_2} = \gamma E_{\gamma_2}^* (1 - \beta \cos \theta^*)$.

γ_1 narrow along x, γ_2 spread along y

4 Chapter 7: Tools For Calculation

4.1 Definitions

4.2 Summary

Polarisation:

Physical polarisations satisfy the transversality condition

$$p_\mu \varepsilon^\mu = 0 \quad (4.1)$$

4.3 Key Formulae

Total cross section

$$\sigma(A + B \rightarrow 1 + \dots + n) = \int d^3p_1 \dots d^3p_n \frac{d\sigma}{d^3p_1 d^3p_2 \dots d^3p_n} \quad (4.2)$$

n particle phase space integral

$$\int d\Pi_n = \int \frac{d^3p_1}{(2\pi)^3 2E_1} \dots \frac{d^3p_n}{(2\pi)^3 2E_n} (2\pi)^4 \delta^{(4)} \left(P^\mu - \sum_j p_j^\mu \right) \quad (4.3)$$

Fermi Golden Rule for a particle width to an n -particle final state f is

$$\Gamma(A \rightarrow f) = \frac{1}{2M_A} \int d\Pi_n |\mathcal{M}(A \rightarrow f)|^2 \quad (4.4)$$

Cross section

$$\sigma(A + B \rightarrow f) = \sum_{\text{spin}} \frac{1}{2E_A 2E_B |v_A - v_B|} \int d\Pi_n |\mathcal{M}(A + B \rightarrow f)|^2 \quad (4.5)$$

4.4 Key Plot and Experimental Signature

Dalitz Plot

4.5 Outline of a Paramount Derivation

2 particle phase space

$$\int d\Pi_2 = \int \frac{d^3p_1}{(2\pi)^3 2E_1} \frac{d^3p_2}{(2\pi)^3 2E_2} (2\pi)^4 \delta^{(4)} \left(P^\mu - \sum_j p_j^\mu \right) \quad (4.6)$$

CM frame

$$\int d\Pi_2 = \frac{1}{8\pi} \int \frac{d\Omega}{4\pi} \quad (4.7)$$

3 particle phase space

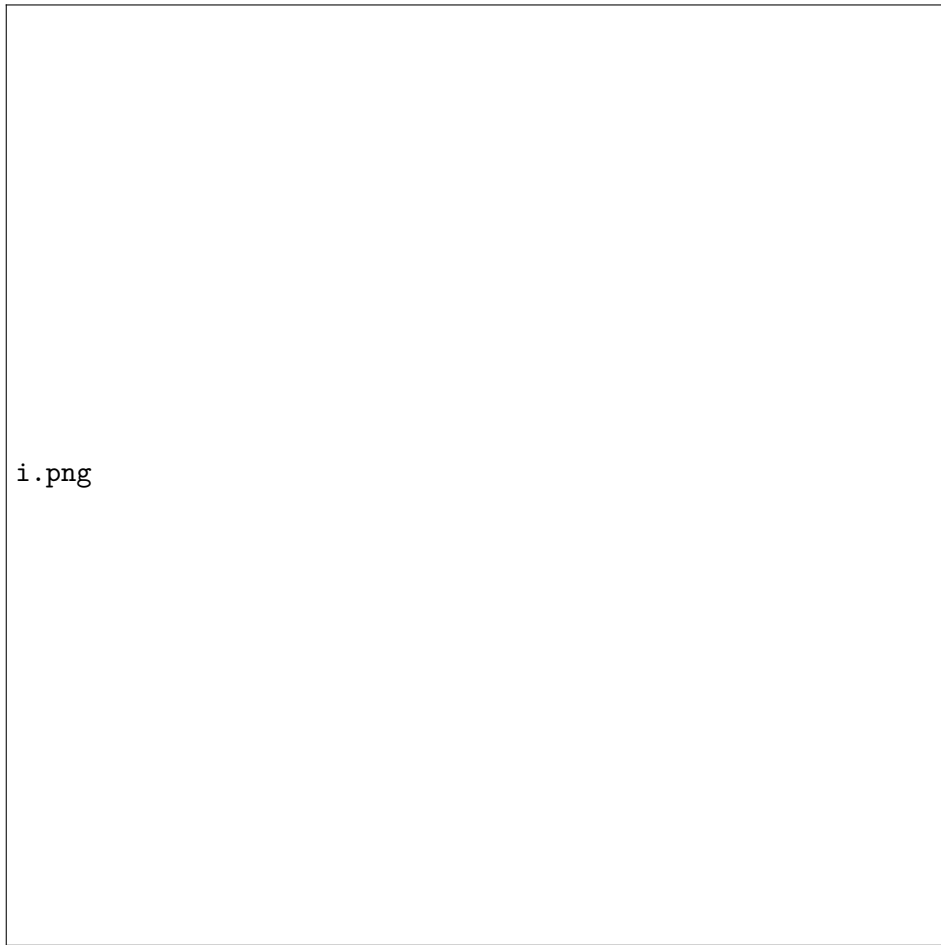


Figure 4.1: Dalitz plot ex 7.2

4.6 Assumptions and Limiting Cases

4.7 Miscellaneous

5 Chapter 8: Electron– Positron Annihilation

5.1 Definitions

Definition 5.1.1 (Helicity). Spin projected along the direction of motion

$$h = \hat{p} \cdot \vec{S} \quad (5.1)$$

Definition 5.1.2 (Chirality Conservation). e_R^- can scatter into an e_R^- or annihilate an e_L^+ , but can't do the inverse. These rules are chirality conservation. chirality is the same as helicity in the massless limit.

5.2 Summary

$e^+e^- \rightarrow \mu^+\mu^-$, the e^+e^- state is annihilated by EM current, current couples to state of EM excitation, then couples to another current matrix element corresponding to creation of muon pair.

$$\mathcal{M}(e^+e^- \rightarrow \mu^+\mu^-) = (-e) \left\langle \mu^+\mu^- \left| j^\mu \right| 0 \right\rangle \frac{1}{q^2} (-e) \left\langle 0 \left| j_\mu \right| e^+e^- \right\rangle \quad (5.2)$$

To evaluate assume high E , $\implies$ masses are negligible. Thus Dirac equation is $i\gamma^\mu \partial_\mu \psi = 0$, where $\psi = \begin{bmatrix} \psi_L \\ \psi_R \end{bmatrix}$ so we interpret the existence of different species of electrons (different chiralities) e_L^- and e_R^- which are independent under EM interactions.

5.3 Key Formulae

$$\gamma^0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \gamma^i = \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \quad \gamma^\mu = \begin{bmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{bmatrix} \quad (5.3)$$

where $\sigma^\mu = (1, \vec{\sigma})^\mu$ and $\bar{\sigma}^\mu = (1, -\vec{\sigma})^\mu$. The creation/ destruction of particles via Dirac fields

$$\begin{aligned} \left\langle 0 \left| \psi_R(x) \right| e_R^-(p) \right\rangle &= u_R(p) e^{-ip_\mu x^\mu} \\ \left\langle e_L^+(p) \left| \psi_R(x) \right| 0 \right\rangle &= v_L(p) e^{+ip_\mu x^\mu}. \end{aligned}$$

The form of Dirac fields

destruction		creation
$e_R^-:$	$u_R(p) = \sqrt{2E} \xi_+$	$e_L^+:$
$e_L^-:$	$u_L(p) = \sqrt{2E} \xi_-$	$e_R^+:$
		$v_L(p) = \sqrt{2E} \xi_+$
		$v_R(p) = \sqrt{2E} \xi_-$

Figure 5.1: Creation and Destruction of electrons and positrons

$$\psi_R(x^\mu) = \int_{\vec{p}} [a_R(\vec{p}) u_R(p^\mu) e^{-ip_\mu x^\mu} + b_L^+(\vec{p}) v_L(p^\mu) e^{+ip_\mu x^\mu}] \quad (5.4)$$

- 5.4 Key Plot and Experimental Signature
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